$^{34}S(^{3}He,d)$ 2017Gi09,1969Gr23

$J^{\pi}=0^{+}$ for ^{34}S g.s.

2017Gi09: a 20-MeV $^{3}He^{2+}$ beam was produced from the MP Van de Graaff tandem accelerator at the Maier-Leibnitz Laboratorium (MLL) and impinged on a 50-μg/cm^2 Ag_2S target 99.999% enriched in ^{34}S with a 8-μg/cm^2 carbon backing. Reaction products were momentum-separated by the Q3D magnetic spectrograph and focused on focal plane detectors consisting of two multiwire proportional counters (MWPCs) and a plastic scintillator. Deduced 22 levels, L-transfers, J, π, spectroscopic factors from FRESCO-DWBA analysis, and resonance strengths. Deduced the thermonuclear $^{34}S(p,\gamma)^{35}Cl$ reaction rate through Monte Carlo methods using the STARLIB code. Calculated the $^{34}S/^{32}S$ isotopic ratio using the hydrodynamical nova model SHIVA.

1969Gr23: a 11.0-MeV ^{3}He beam was produced from the MIT-ONR Van de Graaff generator. The target was 9.3-μg/cm^2 ^{34}S made by evaporating PbS (67.6%^{34}S) onto a formvar backing. Deuterons were detected on 50 μm nuclear emulsions in the MIT multiple-gap spectrograph at 23 different angles from 7.5° to 172.5°. Measured $\sigma(E_d,\theta)$. Deduced 17 levels, J, π, L-transfers, spectroscopic factors from zero-range JULIE-DWBA analysis.

1970Mo01: a 13-MeV ^{3}He beam was produced from the Argonne tandem Van de Graaff accelerator. The target was 200-μg/cm^2 86% enriched ^{34}S. Reaction products were detected using two telescopes of silicon surface-barrier detectors with FWHM≈80 keV and analyzed by a magnetic spectrograph with FHWM≈20 keV at θ<20°. Measured $\sigma(E_d,\theta)$, $\theta_{c.m.}$≈5° to 100°. Deduced levels, J, π, L-transfers, spectroscopic factors from local, zero-range JULIE-DWBA analysis.

1994Ve04: a 25-MeV ^{3}He beam was produced from the Orsay MP Tandem Van de Graaff accelerator. The target was KCl with natural chlorine. Reaction products were momentum-analyzed by an Enge split-pole magnetic spectrograph and detected using a 50-cm long telescope of a position-sensitive proportional counter, proportional ΔE counter, and a plastic scintillator. Measured $\sigma(E_d,\theta)$. Deduced J, π, L-transfer, and spectroscopic factor for the ground state of ^{35}Cl from local, zero-range DWUCK-DWBA analysis.

^{35}Cl Levels

For levels below 6 MeV, spectroscopic factor $(2J+1)C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N is a normalization factor. J is the spin of the populated ^{35}Cl level. N=4.42 is used in 1969Gr23 and 1970Mo01 from 1966Ba54. N=4.43 is used in 1994Ve04. For levels above 6 MeV, C^2S values are from 2017Gi09 without using the factor N. 2017Gi09 also states that it is not possible to distinguish between J=L+1/2 or J=L−1/2 coupling given the experimental uncertainties.

Values of resonance strengths $\omega\gamma$(min) and $\omega\gamma$(max) are calculated by 2017Gi09 using their measured C^2S and calculated Γ_p, assuming the Γ_{tot} is dominated by Γ_γ and thus $\omega\gamma$ depends only on Γ_p.

E(level)†	L‡	Comments
0	2	S: $(2J+1)C^2S$=2.88 in 1969Gr23 for $J^{\pi}=3/2^{+}$; C^2S=1.3 in 1970Mo01 for $J^{\pi}=3/2^{+}$; C^2S=1.07 in 1994Ve04 for $J^{\pi}=3/2^{+}$.
1220 *10*	0	E(level): other: 1220 *12* (1970Mo01). S: $(2J+1)C^2S$=0.26 in 1969Gr23 for $J^{\pi}=1/2^{+}$; C^2S=0.28 in 1970Mo01 for $J^{\pi}=1/2^{+}$.
1758 *10*		
2645 *10*		
2686 *10*	2	S: $(2J+1)C^2S$=0.06 in 1969Gr23 for $J^{\pi}=(3/2,5/2)^{+}$.
3002 *10*	2	E(level): weighted average of 2997 *10* (1969Gr23) and 3010 *12* (1970Mo01). S: $(2J+1)C^2S$=0.14 in 1969Gr23 for $J^{\pi}=(5/2)^{+}$; C^2S=0.04 in 1970Mo01 for $J^{\pi}=5/2^{+}$.
3158 *10*	3	E(level): weighted average of 3156 *10* (1969Gr23) and 3160 *12* (1970Mo01). S: $(2J+1)C^2S$=2.88 in 1969Gr23 for $J^{\pi}=7/2^{-}$; C^2S=0.66 in 1970Mo01 for $J^{\pi}=7/2^{-}$.
3908 *10*		
3962 *10*	0	E(level): weighted average of 3956 *10* (1969Gr23) and 3970 *12* (1970Mo01). S: $(2J+1)C^2S$=0.03 in 1969Gr23 for $J^{\pi}=1/2^{+}$; C^2S=0.06 in 1970Mo01 for $J^{\pi}=1/2^{+}$.
4053 *10*	1	E(level): weighted average of 4048 *10* (1969Gr23) and 4060 *12* (1970Mo01). S: $(2J+1)C^2S$=0.38 in 1969Gr23 for $J^{\pi}=(3/2)^{-}$; C^2S=0.15 in 1970Mo01 for $J^{\pi}=(3/2)^{-}$.
4167 *10*	1	E(level): weighted average of 4165 *10* (1969Gr23) and 4170 *12* (1970Mo01). S: $(2J+1)C^2S$=0.85 in 1969Gr23 for $J^{\pi}=(3/2)^{-}$; C^2S=0.28, 0.62 in 1970Mo01 for $J^{\pi}=(3/2)^{-}$.
5010 *15*		
5163 *15*	(2)	S: $(2J+1)C^2S$=0.11 without specifying J (1969Gr23).
5409 *12*	1	E(level): weighted average of 5407 *15* (1969Gr23) and 5410 *12* (1970Mo01). L: 1 in 1969Gr23 and L=(1,2) in 1970Mo01.

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^{34}S(^{3}He,d) 2017Gi09,1969Gr23 (continued)

^{35}Cl Levels (continued)

E(level)†	L‡	Comments
		S: (2J+1)C^{2}S=0.07 in 1969Gr23 for J^π=(3/2)$^-$.
5660 *15*	2	E(level): 5670 multiplet reported in 1970Mo01. L: 2 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01. S: (2J+1)C^{2}S=0.47 in 1969Gr23 for J^π=3/2$^+$.
5684 *12*	1	E(level): 5670 multiplet reported in 1970Mo01. L: 1 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01. S: (2J+1)C^{2}S=0.11 in 1969Gr23 for J^π=(3/2)$^-$.
5760 *12*	(0,1)	E(level): weighted average of 5760 *15* (1969Gr23) and 5760 *12* (1970Mo01). L: 1 in 1969Gr23 and L=0 in 1970Mo01. S: (2J+1)C^{2}S=0.12 in 1969Gr23 for J^π=(3/2)$^-$; C^{2}S=0.21 in 1970Mo01 for J^π=1/2$^+$.
6284 *4*	(2)	S: (2J+1)C^{2}S=0.086 *2*.
6329 *4*	(0,1)	S: (2J+1)C^{2}S=0.0030 *2* for L=0, 0.0044 *4* for L=1.
6377 *2*	(2,3)	S: (2J+1)C^{2}S=0.0012 *6* for L=2, 0.0024 *8* for L=3.
6427 *2*	(3)	S: (2J+1)C^{2}S=0.049 *2*. $\omega\gamma$(min)=6.5×10^{-24} eV *2*, $\omega\gamma$(max)=2.2×10^{-22} eV *1*.
6468 *2*	(1)	S: (2J+1)C^{2}S=0.034 *1*. $\omega\gamma$(min)=4.2×10^{-14} eV *5*, $\omega\gamma$(max)=4.7×10^{-14} eV *5*.
6491 *2*	(2)	S: (2J+1)C^{2}S=0.072 *2*. $\omega\gamma$(min)=3.7×10^{-13} eV *1*, $\omega\gamma$(max)=4.1×10^{-13} eV *1*.
6545 *2*	(0,1)	S: (2J+1)C^{2}S=0.004 *1* for L=0, 0.0028 *4* for L=1. $\omega\gamma$(min)=7.4×10^{-10} eV *13*, $\omega\gamma$(max)=4.1×10^{-9} eV *1*.
6643 *2*	(1)	S: (2J+1)C^{2}S=0.0144 *8*. $\omega\gamma$(min)=8.5×10^{-6} eV *10*, $\omega\gamma$(max)=8.9×10^{-6} eV *10*.
6674 *2*	(1,2,3)	S: (2J+1)C^{2}S=0.0020 *4* for L=1, 0.0048 *6* for L=2, 0.008 *1* for L=3. $\omega\gamma$(min)=2.4×10^{-8} eV *2*, $\omega\gamma$(max)=4.1×10^{-6} eV *8*.
6761 *2*	(0,1)	S: (2J+1)C^{2}S=0.0056 *12* for L=0, 0.0032 *4* for L=1. $\omega\gamma$(min)=2.4×10^{-4} eV *3*, $\omega\gamma$(max)=1.5×10^{-3} eV *1*.
6778 *2*	(1)	S: (2J+1)C^{2}S=0.0084 *8*. $\omega\gamma$(min)=9.4×10^{-4} eV *12*, $\omega\gamma$(max)=1.0×10^{-3} eV *1*.
6823 *2*	(1)	S: (2J+1)C^{2}S=0.006 *4*. $\omega\gamma$(min)=2.7×10^{-3} eV *3*, $\omega\gamma$(max)=2.7×10^{-3} eV *3*.
6842 *2*	(2,3)	S: (2J+1)C^{2}S=0.0216 *6* for L=2, 0.035 *2* for L=3. $\omega\gamma$(min)=3.3×10^{-5} eV *1*, $\omega\gamma$(max)=7.0×10^{-4} eV *3*.
6866 *2*	(0,2)	S: (2J+1)C^{2}S=0.0160 *2*(L=0)+0.0660 *6* (L=2). $\omega\gamma$(min)=3.0×10^{-2} eV *3*, $\omega\gamma$(max)=4.7×10^{-2} eV *5*.
7066 *2*	(1,2)	S: (2J+1)C^{2}S=0.0088 *8* for L=1, 0.012 *1* for L=2. $\omega\gamma$(min)=2.2×10^{-2} eV *1*, $\omega\gamma$(max)=2.2×10^{-2} eV *1*.
7103 *2*	(1,3)	S: (2J+1)C^{2}S=0.0184 *12* for L=1, 0.025 *2* for L=3. $\omega\gamma$(min)=6.0×10^{-2} eV *3*, $\omega\gamma$(max)=1.30 eV *7*.
7178 *2*	(2)	S: (2J+1)C^{2}S=0.032 *3*.
7194 *2*		
7227 *2*	(0,1)	S: (2J+1)C^{2}S=0.065 *15* for L=0, 0.044 *2* for L=1.
7273 *2*	(0,1)	S: (2J+1)C^{2}S=0.050 *12* for L=0, 0.030 *1* for L=1. $\omega\gamma$(min)=11 eV *2*, $\omega\gamma$(max)=34 eV *3*.
7361 *2*	(1)	S: (2J+1)C^{2}S=0.0040 *8*. $\omega\gamma$(min)=0.30 eV *4*, $\omega\gamma$(max)=3.0 eV *6*.
7398 *2*	(2,3)	S: (2J+1)C^{2}S=0.042 *2* for L=2, 0.0656 *24* for L=3. $\omega\gamma$(min)=0.18 eV *1*, $\omega\gamma$(max)=0.19 eV *1*.

† From 1969Gr23 for E(level)<6 MeV and from 2017Gi09 for E(level)>6 MeV, unless otherwise noted.

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$^{34}S(^{3}He,d)$ 2017Gi09,1969Gr23 (continued)

^{35}Cl Levels (continued)

‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1969Gr23 for E(level)<6 MeV and in 2017Gi09 for E(level)>6 MeV, unless otherwise noted. Firm assignments by 2017Gi09 are placed in parentheses by the evaluators because they are based on very few data points in a very narrow range of angles in the DWBA fit which may not yield definitive and firm results.